

Ch4 Classification

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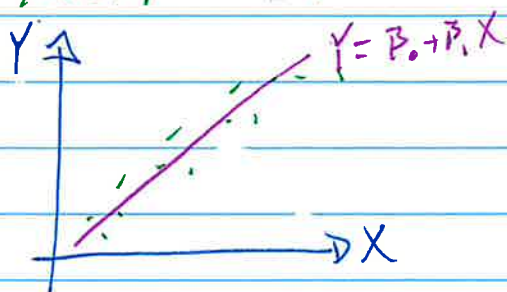
Pg 1

2 main categories in supervised learning:

Ch3 Regression vs Classification

$$Y = f(X) + \epsilon$$

quantity $-\infty < Y < \infty$



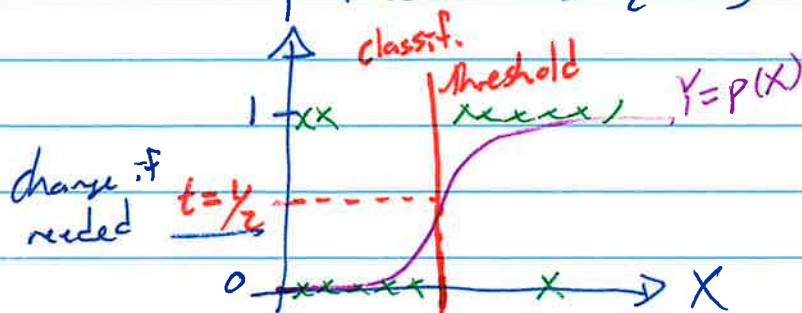
Y finite list of categories without order

$$Y \text{ in } \{Y_1, Y_2, \dots, Y_k\}$$

categories

Basic case: $Y \text{ in } \{Y_1, Y_2\}$

Y recoded as $\{0, 1\}$



$$Y \text{ model by } P_r(Y=1|X) = p(X)$$

Ch4 focuses on how to model conditional probability, $P_r(Y=1|X)$

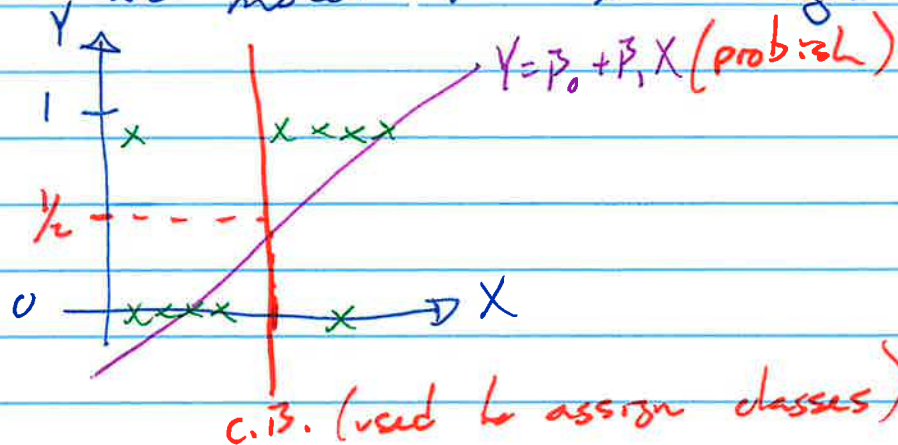
many prob tools

$$0 \leq p(X) \leq 1$$

Use to assign category.

i.e. If $p(X) \geq 1/2$, $Y \rightarrow 1$
else $p(X) < 1/2$, $Y \rightarrow 0$

What if we model via linear regression?



Issues: ① Line $-\infty < \beta_0 + \beta_1 X < \infty$, diff from $0 \leq P(Y=1|X) \leq 1$.

Cancer pred.

Classification
"accuracy" $\rightarrow$
variance.
next time.

② Lines skew to imbalanced classes gives low accuracy in general

③ Gt analysis fails

- ϵ_i 's not normally distr.
- error variance not const
- β coeffs not meaningful, trustworthiness & p-values not useful.

4 diagn
plots
crazy

Instead, model $P(Y=1|X)$ directly.

Ch 4

Many ways: ① Logistic reg (classification prototype)

Bayes rule $\nwarrow$ ② Discriminant analysis (LDA, QDA)

btw
classes

$\rightarrow$ Naive Bayes

linear $\nearrow$ quadr.
decision boundaries

③ Generalized linear model (skip)

④ KNN

Logistic Regression

Model $P(Y=1|X)$

$$p(x) = P(Y=1|X)$$

sigmoid

low → 1
MP8
Class

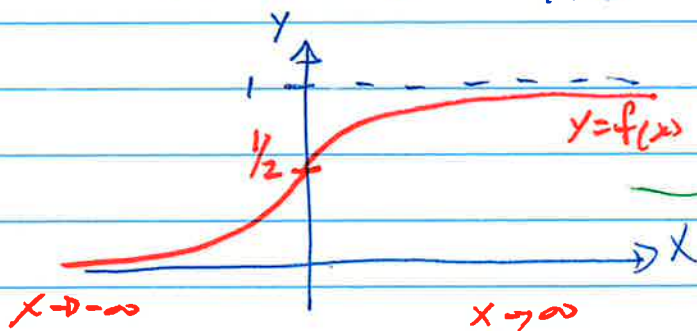
high → 0

hp

Idea: Fit a logistic curve to train data.

Logistic curve: $f(x) = \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x}$

Pop growth
w/ carry
capacity.



fit to data.

Trans data
to
use linear
regression

How? ① $P = p(x) = P(Y=1|X)$ is a probability.

$$0 \leq P \leq 1$$

success prob

Ex | roll 6.

② odds: odds = $\frac{P}{1-P}$

ratio

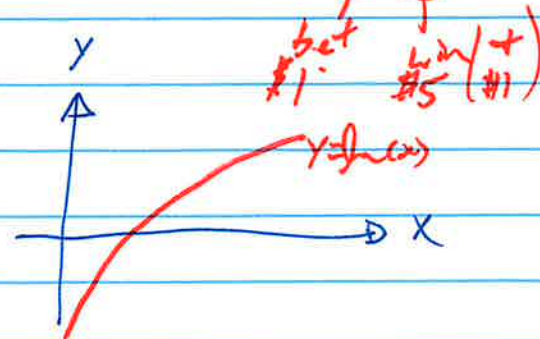
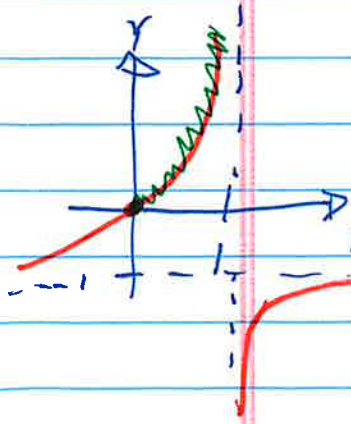
$P = 1/6$
odds = $\frac{P}{1-P} = \frac{1/6}{1-1/6} = \frac{1/6}{5/6} = \frac{1}{5}$ as 1:5

$0 \leq \text{odds} = \frac{P}{1-P} < \infty$

fail prob

③ Log odds: $\ln\left(\frac{P}{1-P}\right)$
 $-\infty < \ln\left(\frac{P}{1-P}\right) < \infty$

math props



④ $\ln\left(\frac{p}{1-p}\right)$ model as linear.

Start
assum.

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

Linear
regression.

Solve for $p = p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$ 😊

(sigmoid
fun)

Notes: ① $\ln\left(\frac{p}{1-p}\right) = \log\text{it}(p)$

② Also write $p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$

③ β_1 coef has nice interpretation (i)

(i) Increase X by 1,

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X \rightarrow \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 (X+1)$$

log-odds increase by β_1

(ii) $\ln\left(\frac{p}{1-p}\right) = \beta_1 \rightarrow \frac{p}{1-p} = e^{\beta_1}$
↑
incr
by

odds
increase

How to fit a logistic regression model?

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X$$

p unknown estimate to $\hat{\beta}_0, \hat{\beta}_1$ (cannot use least squares)

Idea: Choose β_0, β_1 which maximizes the likelihood that data is correct.

Coin flip
Bernoulli
likelihood

$$l(\beta_0, \beta_1) = l((\beta_0, \beta_1) | \{Y_i, X_i\}) \quad (\text{joint prob})$$

how likely β_0, β_1 fits data.

$$= P((Y_1, X_1) \cap (Y_2, X_2) \cap \dots \cap (Y_n, X_n))$$

Assume indep

$$= P(Y_1 | X_1) \cdot P(Y_2 | X_2) \cdot \dots \cdot P(Y_n | X_n)$$

Assume log-odds model

$$= \prod_{Y_i=1} P(X_i) \prod_{Y_i=0} (1 - P(X_i))$$

$P(X)$ products, $P(X) = \text{Pr}(Y=1 | X)$

cases β_0, β_1

Use numerical optimization.

log-likelihood: $\ln(l(\beta_0, \beta_1)) = \sum_{i=1}^n [Y_i \ln(P(X_i)) + (1 - Y_i) \ln(1 - P(X_i))]$

Better to maximize

$$\ln(xy) = \ln(x) + \ln(y)$$

Results in $(\hat{\beta}_0, \hat{\beta}_1)$ best fit solution.

① P-predictors: $x_1, x_2, \dots, x_p$ PS 6
2 Key extensions: ② K72 classes

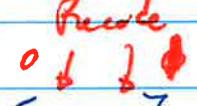
Next time!

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Ch4 Classification

- 3 goals: ① Extend logistic reg- to $X_1, X_2, \dots, X_p$ ^{PS!} 
 $Y \in \{C_1, C_2, \dots, C_K\}$
 ② Other "classic" classifiers (Discriminant analysis)
 ↳ fuzzy feel, skip details ^{Bayes rule}

③ Accuracy metrics

^{st. dev.} (Y, X) , $Y \in \{C_1, C_2\}$ ^{Recall} 

Recap: Logistic regression (simple)

Idea: Model $P(Y=1|X) = p(x)$ ^{prob} _{fcn}

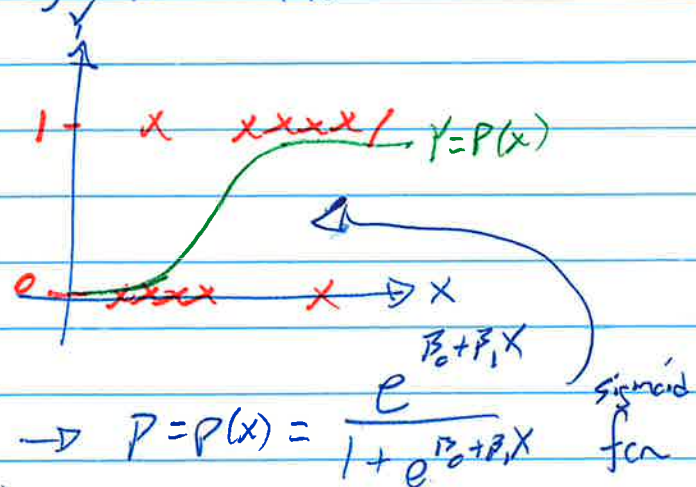
Prob $P = p(x)$

Odds $\frac{P}{1-P}$

Log odds

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

(β_0, β_1)



train
data
likel.

Max likelihood estimate: $l(\beta_0, \beta_1) = \sum_{x_i=1} T[x_i] \ln(p_i) + \sum_{x_i=0} T[(1-p_k)] \ln(1-p_k)$

Numerical: $\ln(l(\beta_0, \beta_1)) = \sum [x_i \ln(p_i) + (1-x_i) \ln(1-p_i)]$

$(\hat{\beta}_0, \hat{\beta}_1) = \arg \max_{(\beta_0, \beta_1)} \ln(l(\beta_0, \beta_1))$

Logistic Reg Extensions:

① $p=1$ predictors: $X_1, X_2, \dots, X_p$

log-odds increase per unit increase X_i

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

$$\hookrightarrow p = P(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}} = \frac{e}{1 + e^{\vec{\beta}^T \vec{X}}}$$

$$\vec{\beta} = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}$$

$$\vec{X} = \begin{bmatrix} 1 \\ X_1 \\ \vdots \\ X_p \end{bmatrix}$$

$$\vec{\beta}^T \vec{X}$$

Get $\hat{\beta}_i$'s by max log likelihood.

②, $k \geq 2$ classes: $Y \in \{Y_1, Y_2, \dots, Y_k\}$ (approaches)

VI: K as reference class

Repeat ideas above for $i=1, 2, \dots, k-1$

$$P_i(x) = P(Y=Y_i | X) = \frac{e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}{1 + \sum_{i=1}^{k-1} e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}$$

Learn from: ① Log odds as LR

② $\sum P_i(x)$ & ref get 1.

IV2 No reference class

$$P_i(x) = P(Y=Y_i | X) = \frac{e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}{\sum_{i=1}^k e^{\beta_{i0} + \beta_{i1} X_1 + \dots + \beta_{ip} X_p}}$$

KS
Soft
(max)

Same result
 β_i 's
means diff

Separate classes

Discriminant Analysis

Classic
Stat
history

1 predictor X
2 classes $Y \in \{0, 1\}$

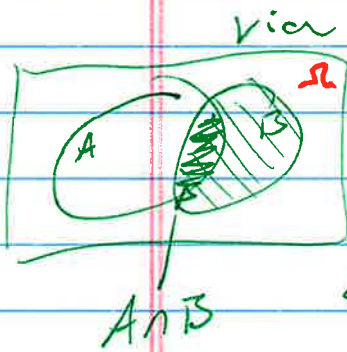
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Goal same: Model $P(Y=1|X) = P(x)$.

Apply Bayes' rule/thm & fit a distribution
to data X

Recap: B.R.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\left(\frac{|A \cap B|}{|B|} = \frac{|A \cap B|/|U|}{|B|/|U|} \right)$$

$$P(B)P(A|B) = P(A \cap B)$$

also

$$P(A)P(B|A) = P(A \cap B)$$

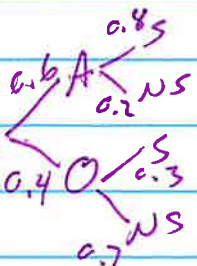
$$P(B)P(A|B) = P(A)P(B|A)$$

How is B.R. useful?

① $P(A|B)$ & $P(B|A)$ connection

② Info update

Apple/oranges
shiny/not



$$P(A|B)$$

posterior
prob updates

likelihood

$$P(B|A)P(A)$$

sh

Normalize
constant

prior

Already
knew

BR for classification

$$\begin{aligned}
 P(Y=1|X) &\stackrel{\text{BR}}{=} \frac{P(X|Y=1) P(Y=1)}{P(X|Y=0) P(Y=0) + P(X|Y=1) P(Y=1)} \\
 &= \frac{P(X|Y=1) P(Y=1)}{P(X) \left(\frac{P(Y=0)}{P(Y=1)} + 1 \right)}
 \end{aligned}$$

(sad face) $P(X|Y=1)$ (sad face) $P(Y=1)$ (happy face) $P(X)$ (happy face) $P(Y=0)$ (sad face) $P(Y=1)$ (happy face)

expand % class 1

D.A. models $P(X|Y=i)$ via normal distr

$$P(X|Y=i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X-\mu_i)^2}{2\sigma^2}}$$

$N(\bar{X}_i, \sigma)$

Gaussian formula.
 Linear
 (sad face) σ^2

LDA, each class
 has same σ
 and GDA allows for
 diff σ_i